Deokhyun Ahn

Ph.D. Candidate, Brain & Cognitive Engineering
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Expected graduation: August 2027 | Available for employment from March 2027

Research Interests

Reliable and efficient video understanding for social good.

Research for social good has always been my first priority, and it will continue to guide my work. I am a Ph.D. candidate in Brain & Cognitive Engineering at Korea University and a student researcher at KIST AI · Robotics.

My early work on sensing applications, including an ultrasonic SLAM helmet for firefighters and indoor drone tracking, shaped my commitment to dependable systems for real-world needs. I now study reliable and efficient video understanding: identifying anomalous behavior, understanding its temporal context, and explaining it with visual evidence.

My work connects video anomaly detection and understanding, adaptive video sampling, action recognition, and vision-language models (VLMs). Through public-safety research at KIST, I work on contextual anomaly detection and behavior analysis for real-world surveillance video.

Recent work includes BUSTER for VLM-guided anomaly detection with adaptive sampling, Where and What for contextual anomaly detection, and Look Closer for action-guided proficiency estimation.

Current focus: why vision-language models hallucinate in video anomaly understanding, and how selective visual verification can make their explanations more faithful.

Research keywords: Video understanding; Video anomaly detection & understanding; Adaptive video sampling; Efficient vision-language models; VLM hallucination & reliability; Temporal & contextual reasoning; Action recognition & human motion analysis; Video surveillance & public safety.

Education

Korea University Mar 2023 – Present
Ph.D. Candidate, Brain & Cognitive Engineering *Seoul, KR*

- Expected graduation: August 2027.
- Study contextual video understanding inspired by human cognitive processes.

Kwangwoon University Seoul, KR
M.Eng. in Electronic Engineering *February 2020*
B.Eng. in Electronic Engineering *August 2015*

- Thesis: *A Study on Passive Target Tracking in Three-Dimensional Indoor Environment by Using Deep Neural Network.*
- Completed 19 months of mandatory military service.

Research & Work Experience

Student Researcher Jan 2023 – Present
KIST AI · Robotics Institute / Vision Intelligence Group *Seoul, KR*

- Video anomaly understanding; Surveillance video; Vision language models.
- Develop public safety video systems spanning action recognition, contextual anomaly detection, and vision language models that explain anomalous behavior. [1] [2] [3] [4] [5]
- Developed research prototypes and integrated them into a real-world monitoring system.
- Served as frontend team lead for the monitoring system, contributing directly to implementation and coordinating the team's development work.

Research Intern Jan 2022 – Feb 2023

KIST AI · Robotics Institute | Vision Intelligence Group *Seoul, KR*

- Began surveillance video research through urban public safety projects and national research programs.

Researcher Oct 2021 – Dec 2021

The Korea Transport Institute *Sejong, KR*

- Contributed to autonomous bus development through SLAM, control center systems, policy research, and coordination.

Selected Publications

[1] **BUSTER: Adaptive Sampling for VLM-Guided Unsupervised Video Anomaly Detection.**
Ahn D., Choi Y., Han J.-H., Kim I.-J., Kim H.
British Machine Vision Conference (BMVC), 2026 | Accepted

[2] **Learning Robust Representations for Few-Shot Action Recognition with Frame-Level Ambiguities.**
Ahn D., Jo Y., Lee U.-S., Park H., Han J.-H., Kim H.
IEEE Access, 2026 | Accepted

[3] **Look Closer: Action-Guided Dense Visual Dynamics for Proficiency Estimation.**
Ahn D., Kim H., Han J.-H., Ham B., Choi H., Kim I.-J., Kim H.
CVPRW 2026 | Multimodal Human Motion Analysis | Accepted; Pitch and Poster Presentation

[4] **Where and What: Contextual Dynamics-Aware Anomaly Detection in Surveillance Videos.**
Ahn D., Jo Y., Kim D., Nam G. P., Han J.-H., Kim H.
IEEE Transactions on Image Processing, 2025 | Published
JIF 15.3 | Top 2.17% in E.E. Eng. (2025 JCR)

[5] **A Foundational Research Framework for Real-World Abandoned Object Detection: Train-Free Baseline and a Standardized Benchmark.**
Kim D., **Ahn D.**, Jo Y., Park H., Lee S., Kim H.
Expert Systems with Applications, 2025 | 130658 | Published

[6] **A Passive Tracking System Based on Geometric Constraints in Adaptive Wireless Sensor Networks.**
Zhou B., **Ahn D.**, Lee J., Sun C., Ahmed S., Kim Y.
Sensors, 18(10), 3276, 2018 | Published

[7] **2D Indoor Map Building Scheme Using Ultrasonic Module.**
Ahn D., Kim N. M., Park J. H., Kim Y. O.
J. Korean Inst. Comm. and Info. Sci., 41(8), 986–994, 2016 | Published

Research Grants & Projects

Real-Time Intelligent Surveillance for Public Safety ~$24M

National R&D *Five year program*

- MSIT & MOTIE; with Yonsei University and the Korean National Police Agency.
- Video anomaly detection that uses context across city scale CCTV control centers with about 1,000 cameras, validated in Anyang City.

Mobile Platform for Active Safety Services ~$9.5M

Korean National Police Agency & MOTIE *Five year program*

- Government research program using mobile platforms for proactive public safety services.

Anomaly Detection in Correctional Facilities New initiative

KIST AI · Robotics

- Detecting inmate anomalous behavior in custodial environments.

Awards & Honors

AIR Outstanding Research Award Dec 2025

KIST AI · Robotics Institute

- In recognition of outstanding research achievements during 2025.

Excellent Paper Award, Graduate School Feb 2025

Korea University

- For outstanding research performance and publication of an excellent paper during doctoral studies.

Outstanding Oral Presentation Award Nov 2024

KIST Academia and Research Convergence Conference

- For the presentation "Multi-Aspect Contextual Dynamics for Anomaly Detection in Surveillance Videos."

Division Commander's Commendation Jul 2021

Capital Mechanized Infantry Division

- For a combat development proposal on reconnaissance drone AI improvement and acoustic detection using AI and big data.

Teaching

Senior Capstone Project Sep – Dec 2018

Kwangwoon University | Instructor: Prof. Youngok Kim *Teaching Assistant*

- Assisted in managing and supporting the Capstone Design Project as a TA for 50+ students.

Computer Architecture Mar – Jun 2017

Kwangwoon University | Instructor: Prof. Youngok Kim *Teaching Assistant*

- Graded assignments and provided feedback for 30+ students with Q&A sessions.

Technical Skills

Programming: Python, C, Java, Git, Linux

Machine Learning: PyTorch, TensorFlow, Keras, NumPy, Pandas, Matplotlib

Sensors: Ultrasonic, Beacon, RF, Wi-Fi, 3-axis Accelerometer

Embedded & IoT: Raspberry Pi, Arduino

Updated October 2026